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Transformer based multiple superpixel-instance learning for weakly supervised segmenting lesions of interstitial lung disease

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ABSTRACT

Automatic and accurate segmentation of lesions from high-resolution computed tomography (HRCT) images play a critical role in diagnosis of interstitial lung disease (ILD). Fully supervised methods require considerable expertise annotations, resulting in time-consuming and labor-intensive cost. Contrastively, weakly supervised segmentation (WSS) algorithms could alleviate greatly this issue of medical image annotation. Multiple-instance learning (MIL), a class of weakly supervised learning (WSL), has proven to be effective in image segmentation tasks. However, most existing MIL methods based on patches or pixels lack meaningful edge information of the region of interests (ROIs) and leave out of considering the dependence relations of inter-instances. In this study, we propose a novel Transformer based multiple superpixel-instance learning (MSIL) for pixel-level lesions segmentation of ILD. This method introduces superpixels into a MIL framework, which solves the problem of inconsistency in lesion boundary segmentation caused by using patches as instances. In addition, the proposed method introduces Transformer into the MIL framework to capture global or long-range dependencies and establish the relationship between superpixel instances based on the multi-head self-attention, which solves the shortcoming that instances are independent of each other in MIL. Experimental results on private HRCT images confirm that our proposed method can achieve state-of-the-art (SOTA) performance compared to other WSS algorithms, MIL-based WSS algorithms, CAMs-based classification algorithms (CA) and fully supervised segmentation algorithms (FSS).

1. Introduction

Interstitial lung disease (ILD) refers to a variety of diseases that mainly affect the lung interstitium and disrupt the gas exchange process (Wijsenbeek, Suzuki, & Maher, 2022), which can lead to potentially life-threatening complications (Gao et al., 2018). Early diagnosis is crucial for making treatment decisions. High-resolution computed tomography (HRCT) is generally considered central to the diagnosis of ILDs due to the specific radiation attenuation properties of lung tissue (Raghu et al., 2018). Therefore, HRCT image segmentation is of great significance in computer-aided diagnosis (CAD) and can help radiologists quickly locate lesions. However, radiographic assessment of ILD requires considerable experience and expertise, and can vary widely among experienced radiologists (Walsh et al., 2016). In addition to this, the heavy task of reading images places a heavy workload on radiologists, and the development of automated quantitative methods may be of important clinical relevance and interest. Furthermore, automated methods can overcome interobserver variability in visual assessment of the extent of ILD (Collins et al., 1994).

With the development of deep learning and artificial intelligence, convolutional neural networks (CNNs) and their variants have become the most commonly used methods in medical analysis. Current effective supervision methods always rely on a large number of high-quality annotations. However, obtaining these fine-grained annotations is usually time-consuming and labor-intensive for radiologists. Therefore, in this paper, we propose to use a weakly supervised segmentation (WSS) method with image-level labels to automatically segment the region of interests (ROIs) in HRCT images.

Current research on ILD mainly focuses on lesions classification (Agarwala et al., 2020; Anthimopoulos, Christodoulidis, Ebner, Christe, & Mougiakakou, 2016; Gao et al., 2018; Guo, Xu, & Zhang, 2019; Li, Cai, Wang, Zhou, Feng, & Chen, 2014; Song et al., 2015; Sukanya Doddavarapu, Kande, & Prabhakara Rao, 2020; Vinta, Lakshmi, Safali, &

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Table 1
The summary of different methods for ILD. SSL, WSL, FSL are denoted to semi-supervised learning, weakly supervised learning and fully supervised learning, respectively. LSRE is denoted Locality-constrained Subcluster Representation Ensemble. SKC is denoted Spherical K-means clustering.

Authors	Method	Type of Supervision			Performance Metrics
		SSL	WSL	FSL	
Classification					
Li et al. (2014)	Customized CNNs			✓	0.762 Recall, 0.739 Precision
Song et al. (2015)	LSRE			✓	0.825 Precision, 0.833 F_1
Anthimopoulos et al. (2016)	Customized CNNs			✓	0.865 Accuracy
Gao et al. (2018)	Customized CNNs			✓	0.879 Accuracy, 0.702 F_1
Wang et al. (2019)	SKC + CNN		✓		0.880 Accuracy
Guo et al. (2019)	small kernel DenseNet (SK-DenseNet)			✓	0.984 F_{avg}
Sukanya Doddavarapu et al. (2020)	Customized CNNs			✓	0.947 Accuracy
Agarwala et al. (2020)	FCN			✓	0.653 Sensitivity
Vinta et al. (2024)	MobileUNetV3			✓	0.987 Precision, 0.981 F_1
Object detection					
Agarwala et al. (2021)	GoogleNet			✓	0.678 F_1
Segmentation					
Anthimopoulos et al. (2018)	FCN + dilated Conv	✓			0.818 Accuracy
Cai, Liu, et al. (2023)	Mean-Teacher + Self-training		✓		0.871 DSC
Guo et al. (2023)	SEAM + up-sample + adaptive threshold + augmented losses		✓		0.923 Accuracy, 0.487 mDSC
Classification and Segmentation					
Ours	Transformer + MSIL		✓		0.998 Precision, 0.856 F_1, 0.856 DSC, 0.605 mIoU

Kumar, 2024; Wang et al., 2019) and detection (Agarwala et al., 2021). Vinta et al. (2024) first use UNet++ to segment lung parenchyma, then use Refined Attention Pyramid Network (RAPNet) to extract deep features of lung parenchyma, and finally use MobileUNetV3 to classify ILD. Agarwala et al. (2021) use GoogleNet to automatically detect ILD lesions without doing lung field segmentation. To the best of our knowledge, there are only a few studies on pixel-by-pixel segmentation of ILD lesions in CT images. Anthimopoulos et al. (2018) propose a dilated fully convolutional network for segmenting six typical ILD patterns on the public ILD multimedia database maintained by the University Hospital of Geneva (HUG) (Depeursinge et al., 2012). Cai, Liu, et al. (2023) propose a semi-supervised learning method for segmenting ILD lesions from CT images. This method uses Mean-Teacher as the training model and uses consistency regularization to encourage consistent output of the two models under different perturbations. However, all the above methods require pixel-level annotation. Guo, Abdi, Shinagawa, Jerebko, and Farhand (2023) propose a weakly supervised anomaly segmentation algorithm based on the Self-supervised Equivariant Attention Mechanism (SEAM) model and compose of self-supervision and autovariant attention mechanisms. This method can distinguish lesions of ILD and segment abnormal areas. However, the mean Dice Similarity Coefficient (mDSC) of this method is only 0.4867. Table 1 summarizes the current research on classification and segmentation of ILD lesions.

Multiple-Instance Learning (MIL) (Dietterich, Lathrop, & Lozano-Pérez, 1997) is a subset of weakly supervised methods, which was first proposed by Dietterich et al. for predicting drug activity. Since then, a large number of MIL-based algorithms have been developed. The main idea of the MIL method based on deep learning is to segment an image into multiple patches, each image is regarded as a bag in the MIL, and the patches in the image are regarded as instances in the MIL. The labels of the bags are image-level labels, but the instance labels in the bags are unknown. In addition to bag-level classification, the MIL also can predict instance-level labels. At present, many studies apply MIL to medical image processing (Han et al., 2020; Hashimoto et al., 2020; Safta, Farhangi, Veasey, Amini, & Frigui, 2019). For CT image analysis, Safta et al. (2019) use the MIL to classify lung nodules as benign versus malignant from thoracic CT images. Han et al. (2020) use the MIL to screening of COVID-19 from 3D chest CT images. However, All of these works are currently based on square patch instances, such patch-based MIL can obtain outstanding results for disease classification. However, due to the diffuse of the ILD lesion, patch-based MIL lacks boundary information and cannot accurately segment the lesion. More importantly, these works are based on CNNs backbone to learning the patch instance aimlessly and aggregating them in the last fusion layers, thereby ignoring the global or long dependencies of inter-instance.

In this work, we propose ViT-MSIL (ViT based MSIL), which introduces superpixels into the MIL framework for the first time, specifically for ILD lesion segmentation. Superpixels serve as the basic processing unit in the MIL, where pixels with similar intensity and positions within the same image are grouped into a superpixel as an instance. The use of superpixels places greater emphasis on ROIs boundary information and alleviates the adverse effect compared to inappropriate patch instances or pixel instances. The proposed ViT-MSIL not only solves the difficulty of manual annotation, but also solves the problem of unclear ROIs boundaries in segmentation tasks caused by the MIL as a classification algorithm. In addition, the utilization of Transformer can capture global or long-range dependencies and establish the relationship between superpixel instances based on the multi-head self-attention mechanism, which solves the shortcoming that instances are independent of each other in the MIL. Our proposed ViT-MSIL framework achieves the DSC of 0.856 and mean Intersection-over-Union (mIoU) of 0.605 in segmenting the ILD lesion from the internal HRCT image. Our main contributions can be summarize as follows.

- We propose a novel WSS algorithms (ViT-MSIL), introducing superpixels and transformer into the MIL framework to provide more meaningful edge information and capture long-range dependencies of superpixel instance.
- We explore the combination of Transformer, superpixels and MIL for WSS on HRCT images, which is the first attempt for this task to our the best knowledge.
- Our extensive experimental results show that our method achieves state-of-the-art (SOTA) performance. And from the visualization results, our method can segment the lesion edge more accurately than the patch-based MIL method.

The remainder of this paper is as follows. Section 2 presents related work; Section 3 introduces the overall ViT-MSIL framework and detailed module descriptions.; Section 4 introduces the experimental data and the experimental results; Section 5 introduces what needs to be discussed in the paper. Finally, Section 6 gives a conclusion of the paper.

2. Related works

This section would give a brief review of some works related to our study, including MIL based on CNNs, MIL based on Transformer, and superpixels for medical images segmentation.

2.1. MIL based on CNNs

Most of works about MIL has been focused on the classification of medical images (Campanella et al., 2019; Chikontwe, Kim, Nam, Go, & Park, 2020; Kanavati et al., 2020; Qi et al., 2021; Schirris, Gavves, Nederlof, Horlings, & Teuwen, 2022; Silva-Rodríguez, Schmidt, Sales, Molina, & Naranjo, 2022; Yao, Zhu, Jonnagaddala, Hawkins, & Huang, 2020; Zhang et al., 2022, 2024), there are only a few works that apply MIL to segmentation tasks. Xu, Zhu, Chang, and Tu (2012) propose a histopathological image segmentation algorithm in which the concept of multi-cluster instance learning (MCIL) was introduced. The MCIL algorithm can perform image-level classification, patch-level segmentation, and patch-level clustering simultaneously. However, they adopt a boosting algorithm that specifies all feature types in advance, but the features in our method are learned automatically. Jia, Huang, Chang, and Xu (2017) construct a MIL algorithm for histopathology image segmentation based on deep weak supervision (DWS-MIL), but the classifier in this method need pre-training. Xu et al. (2019) propose CAMEL, which uses MIL to segment histopathology images using image-level labels, and the classifier of this method does not require pre-training, thus increasing the flexibility in choosing the network architecture. Similarly, Lerousseau et al. (2020) use MIL to segment tumors in histopathology images. However, CNN-based MIL will ignore the global or long dependencies between instances, resulting in incomplete feature learning.

2.2. MIL based on transformer

Transformer has shown excellent performance in both natural language processing (NLP) and computer vision (CV) (Vaswani et al., 2017). Unlike CNNs that focus on the local receptive field of each convolutional layer, Transformer can capture global or long-range dependencies through a self-attention mechanism. That is, the self-attention mechanism aggregates contextual information from other instances into a bag in MIL. There have been some studies showing that incorporating Transformer into MIL has better performance than CNNs (Cai, Feng, et al., 2023; Cersovsky, Mohammadi, Kainmueller, & Hoehne, 2023; Li, Yang, et al., 2021; Lu et al., 2021; Naik et al., 2020; Ren et al., 2023; Shao et al., 2021; Yu et al., 2021). Qian et al. (2022) introduce the Transformer into the MIL framework for segmentation of histopathology images for the first time. It combines the Swin Transformer with MIL to capture global or remote dependencies, and introduces deep supervision, overcomes the limitations of labeling in WSL, and makes better use of hierarchical information. However, because the pathological image is very large, the authors compress the image to 256×256, which will cause problems such as pixel compression. Li et al. (2023) introduce a self-attention mechanism into the MIL framework for pixel-level segmentation in histopathology images. Whether CNN-based MIL or Transformer-based MIL, most of the works use patches or pixels as instance, but the lesion is diffuse, using patches as instance will lead to inaccurate edge segmentation of the lesion. However, using pixels as instances will increase the amount of calculation.

2.3. Superpixels for medical images segmentation

Superpixels refer to groups of pixels with similar characteristics in terms of image intensity and spatial location (Ren & Malik, 2003). It has been successfully applied to medical image segmentation tasks based on deep learning. Li et al. (2020) propose a label softening method driven by image context information that introduced the concept of superpixels to improve segmentation performance, especially near the boundaries of different categories. Li, Gao, and He (2021) propose a novel robust learning strategy for semantic image segmentation. This method combines the noise-aware training of the segmentation network guided by superpixels and noise label refinement, which can better utilize the structural prior of the image and the correlation of pixel labels, and effectively mitigate the impact of label noise. Wang, Lyu, and Tang (2023) propose a masked image modeling (MIM) method based on superpixels to improve the segmentation performance of skin lesions and achieved good results. Weng et al. (2024) propose Superpixel and Confident Learning Guide Point Annotations Network (SCLGPA-Net) based on the teacher-student architecture, which can learn OCT fluid segmentation from limited fully-annotated data and abundant point-annotated data.

3. Methodology

In this section, we describe our proposed ViT-MSIL for ILD lesion identification and segmentation, as shown in Fig. 1. Specifically, we first provide a CNN network (named LPSM) to segment lung parenchyma from HRCT, then we use the ViT-MSIL module to identify and segment the lesions. In the following subsections, we would describe all these components in detail.

3.1. Segmentation of lung parenchyma

Due to the three-dimensional tomographic properties of CT, a large number of CT images are generated for each patient during CT examination, especially HRCT. In addition, CT is used to image the whole human body tissues, as shown in Fig. 2, including not only lung parenchyma, but also peripheral tissues such as blood vessels and bronchi, as well as extra-lung tissues such as thorax. If the feature extraction of ILD lesions is directly carried out on the original HRCT images, the workload and calculation time are huge. Furthermore, according to the definition of clinical medicine, ILD is a disease that occurs in the lung cavity. In order to avoid the interference of human tissues outside the lung cavity to the identification of ILD lesions, it is hoped that the ROIs of the study can be focused on the lung cavity, so as to reduce the interference and the amount of calculation of the algorithm.

We extract lung parenchyma based on the method in the previous work (Lai et al., 2022). It is divided into three steps: Firstly, digital conversion of HRCT data is carried out, and then some data is sampled equidistant. The lung parenchyma segmentation model (LPSM) was then used to segment the lung parenchyma. LPSM includes saliency segmentation algorithm and post-processing such as corrosion and expansion. After the LPSM, the noise-removed lung parenchyma region can be obtained for subsequent operations.

3.2. Vit-MSIL module

3.2.1. Instance embedding

Due to the diffuse nature of ILD lesions, simply using patches as instance to segment lesions will lead to blurred edges, so we propose the ViT-MSIL module. In this module, we propose a novel multi-instance learning method. Specifically, we use the Felzenszwalb (Felz) (Felzenszwalb & Huttenlocher, 2004) algorithm to generate superpixels instead of patch-level instance in traditional MIL. We construct an end-to-end

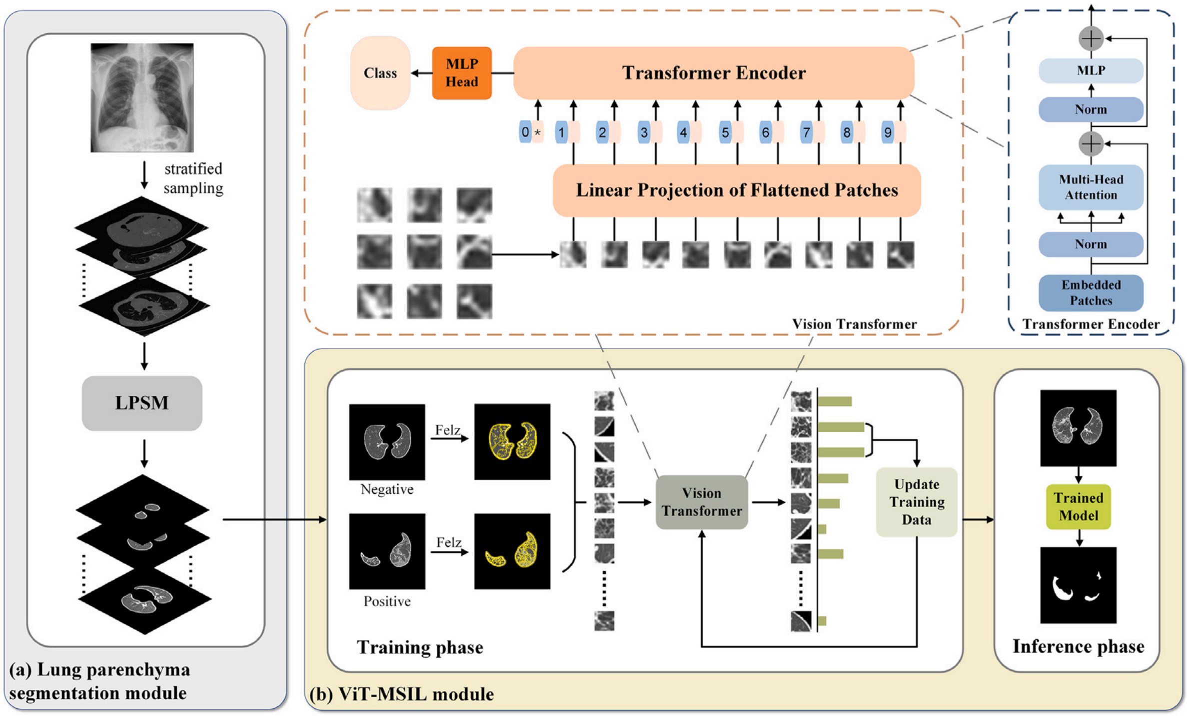


Fig. 1. Overall pipeline of the proposed ViT-MSIL framework. The model is mainly divided into two parts, the first part (a) is lung parenchyma segmentation module (LPSM) which is used to segment the lung parenchyma from HRCT, and the second part (b) is the main VIT-MSIL module, where we combine ViT, superpixel and MIL to classify and segment interstitial lung disease lesions.

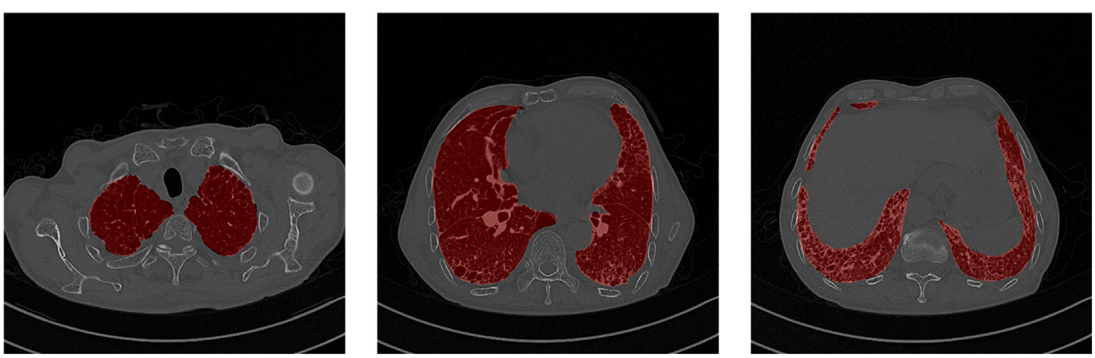


Fig. 2. The lung CT images, the red area is the lung parenchyma, and the rest is the background, including the surrounding tissues, other organs of the abdomen and CT instrument bed.

MIL method to realize the automatic learning of superpixels-level segmentation from image-level labels. As shown in Fig. 3, since superpixels are irregular and difficult to be used as the input of a convolutional network, we first calculate the bounding rectangle of each superpixel, then calculate the average of these rectangles, and then resize the superpixel to the average rectangle before inputting it into the convolutional network for subsequent operations.

Using the transfer learning method, the pre-training model and classifier F_{vit} of vision transformer (ViT) (Dosovitskiy et al., 2020) are first used as the initial parameters and classifier for ViT-MSIL training. In the process of training, the ILD image X is formulated as a bag, and its width, height and channel are W, H and C, respectively. We denote the training dataset by $S = \{(X_i, Y_i), i = 1, 2, 3, \ldots, n\}$, where X_i denotes the image of the ith input and $Y_i \in \{0, 1\}$ represents the image-level label assigned to the ith input image, where $Y_i = 0$ denotes an image without ILD and $Y_i = 1$ denotes an image with ILD. As mentioned earlier, our task is to be able to perform superpixel-level predictions learned from image-level labels, in which case each superpixel is referred to as an instance $x_p \in R^{n\times(H_p \times W_p \cdot C)}$, where the H_p and W_p denote width and height of each instance, respectively. Clearly, the number of instances n can be calculated via $n = W \times H/(H_p \times W_p)$.

Then, following the operations in existing ViT pipelines, we use a transformation matrix E to project each of these instances into a D-dimensional linear representation as Eq. (1).

$$Z_0 = [X_{class}; x_p^1 E; x_p^2 E; \ldots x_p^n E] + E_{pos}, E \in R^{(H_p \times W_p \cdot C)\times D}, E_{pos} \in R^{(n+1)\times D} \tag{1}$$

The input Z_0 is processed by multiple encoder units subsequently, as this part of our ViT-MSIL is still in the ViT backbone. Specifically, assume there are L encoder units in the backbone, and let Z_ℓ and $Z_{\ell-1}$ denote the feature representation before and after processing by the ℓth unit, respectively, where $\ell = 1, 2, \ldots, L$. This step can be represented as Eqs. (2) and (3):

$$Z'_\ell = MSA(LN(Z_{\ell-1})) + Z_{\ell-1} \tag{2}$$

$$Z_\ell = MLP(LN(Z'_\ell)) + Z'_\ell \tag{3}$$

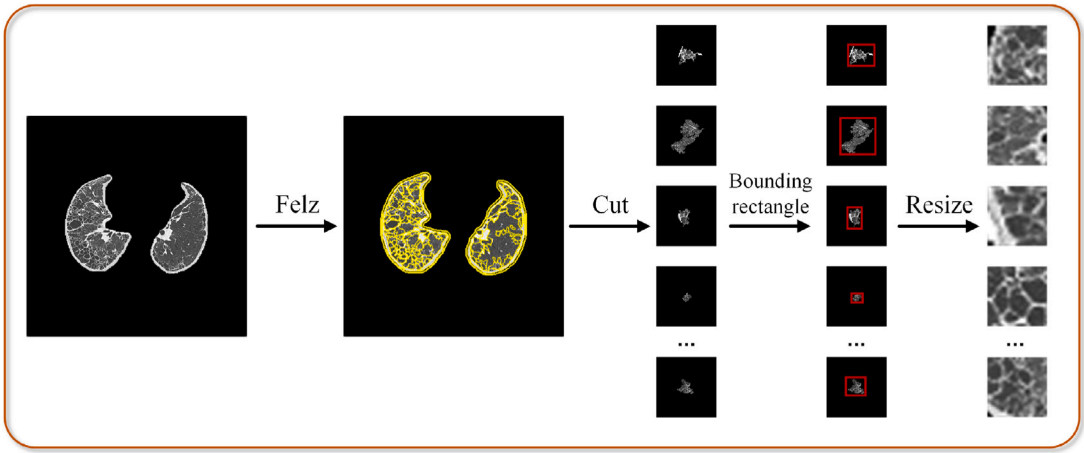


Fig. 3. The process of superpixels extraction by ViT-MSIL Module.

where MSA(·) denotes the multi-head self-attention, LN(·) denotes the layernorm, and MLP(·) denotes the multi-layer perception.

3.2.2. Training of ViT-MSIL

In terms of extracting feature of superpixels, we employ the ViT as backbone to get a superpixel-level classifier F_{vit}. The algorithm is described in Algorithm 1. To be specific, for each bag, we first classify the dataset S using the initial classifier F_{vit} to obtain the classification result for each instance in the bag, we denote by $\hat{Y_{ik}}$ the probability that the kth instance in the ith image is positive, where $k = \{1, 2, \dots .n\}$ and n is the total number of instances of the image. Take positive instances with high probability in each positive bag and negative instances with high probability in each negative bag and append them to the dataset S to update the training data. Finally, S is used to train the classification network to obtain the updated classifier F_{vit}. The network parameters are iteratively fine-tuned by repeating the above three steps to obtain the final instance-level classification network.

In the process of updating classifier F_{vit}, we use the cross-entropy loss function by comparing the output $\hat{Y}_i$ with the target Y_i:

$$L_{MIL} = \sum \left(Y_i log \hat{Y}_i + (1 - Y_i) log(1 - \hat{Y}_i) \right) \tag{4}$$

where Y_i denotes the label of the ith image, $\hat{Y}_i$ denotes the probability that ith image is predicted to be positive.

Algorithm 1: Training of ViT-MSIL.

Input: ILD images $S = \{(X_i, Y_i), Y_i \in (0, 1)\}$, where X_i represents the bag, and Y_i denotes its corresponding class label
Output: Superpixel-level classifier f

Using Felzenszwalb algorithm to generate superpixels. $X_i = (x_{ij}, y_{ij})$, in which x_{ij} denotes instance in bag X_i, y_{ij} denotes the label of x_{ij};
Initialization: superpixels labels $y_{ij} = Y_i$, temporary training dataset S, maximum epoch $E \in \mathbb{R}$;
Extend all x_{ij} into S;
Train a basic classifier F_{vit} via dataset S;
while *True* **do**
  Update y_{ij} via F_{vit};
  Clear up S;
  if X_i *is positive* **then**
    Classify each superpixel via F_{vit} and obtain its probability of being positive;
    Choose superpixel with the highest probability and append it into S;
  end
  else if X_i *is negative* **then**
    Classify each superpixel via F_{vit} and obtain its probability of being negative;
    Choose superpixel with the highest probability and append it into S;
  end
  retrain the classifier F_{vit} via updated S;
  epoch+1;
  if *epoch* == E **then**
    Break;
  end
end
Set f=F_{vit}.

4. Experiments and results

4.1. Dataset description

The ILD patient data used in this paper are all obtained from the radiology department database of Shanxi Bethune Hospital, which do not require informed consent from patients due to its retrospective nature. The study is conducted in accordance with the principles of the Declaration of Helsinki.

The dataset for the study were collected from March 2018 to August 2021, which consists of 306 ILD (positive) and 200 non-ILD (negative) patients. The distribution of dataset is shown in Table 2, and the resolution of each image is 512 × 512.

Based on our previous work (Lai et al., 2024), the ground truth (GT) of the dataset are pixel-level annotations which were provided by two radiologists. (1) If the overlap between the two lesion regions labeled by the two radiologists is larger than 80%, we use the intersection of the two regions as the ground truth; (2) if the overlap is less than 80%, or if a lesion region is annotated by one radiologist but ignored by another, a third senior radiologist will step in to help determine whether the region is diseased or not.

Table 2
Details of dataset distribution.

Division	Type	#Patients	#Images
Train	Positive	150	20 716
	Negative	130	20 260
Test	Positive	156	33 213
	Negative	70	10 200

4.2. Evaluation metrics

We employ precision, recall, F_1-score to evaluate classification performance and DSC, mIoU (Everingham et al., 2015), Hausdorff Distance (HD) (Huttenlocher, Klanderman, & Rucklidge, 1993) to evaluate segmentation performance. DSC and mIoU are the standard metric of semantic segmentation, HD is defined as the maximum distance between the edge of prediction and GT, which is able to evaluate the edge prediction accuracy. Smaller values of HD represent more accurate predictive boundary segmentation. These formulations related to the evaluation metrics are defined in the follows:

$$Precision = \frac{TP}{TP + FP}. \tag{5}$$

$$Recall = \frac{TP}{TP + FN}. \tag{6}$$

where TP, FP, FN are denoted as true positive, false positive and false negative, respectively.

$$F_1 = \frac{2 * Precision * Recall}{Precision + Recall}. \tag{7}$$

$$DSC = \frac{2|S_A \cap S_B|}{|S_A| + |S_B|}, \tag{8}$$

$$mIoU = \frac{1}{n}\sum \frac{|S_A \cap S_B|}{|S_A| + |S_B| - |S_A \cap S_B|}, \tag{9}$$

where S_A denotes the GT and S_B denotes the segmentation results via our method, n denotes the number of classes.

$$HD(A, B) = max(h(A, B), h(B, A)) \tag{10}$$

with

$$h(A, B) = \max_{a\in A}\left\{\min_{b\in B}\|a - b\|\right\}, \tag{11}$$

$$h(B, A) = \max_{b\in B}\left\{\min_{a\in A}\|b - a\|\right\}, \tag{12}$$

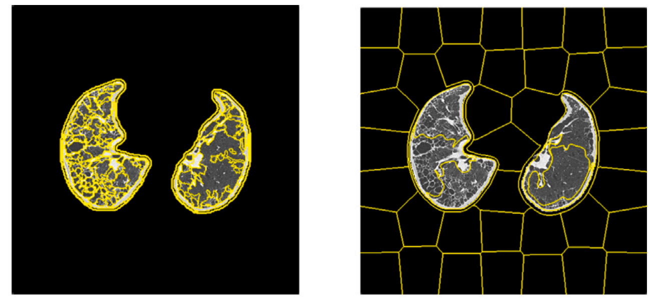
where $\|\cdot\|$ is the normal form of distance between point sets A and B.

4.3. Implementation details

All experiments are implemented on the PyTorch framework and performed on an NVIDIA Tesla V100 32 GB GPU and an Intel(R) Core(TM) i7-8700 @3.2 GHz. We choose Global Context-Aware Progressive Aggregation Network for Salient Object Detection (GCPANet) (Chen, Xu, Cong, & Huang, 2020) as the backbone of LPSM and ViT as the backbone of ViT-MSIL, respectively. For LPSM training, the mini-batch Stochastic gradient descent (SGD) was used to optimize the whole network with the batch size of 32, the momentum of 0.9, and the weight decay of $5e$-4. For ViT-MSIL training, the Adam optimizer was used to train weights with a decay of $1e$-5 and a fixed learning rate of $1e$-5, and batch size is set to 128 and epoch is set to 200 to save the best training weights.

4.4. Ablation experiment

To verify our proposed method and configurations' usefulness, we perform ablation experiments. The ablation experiments are mainly divided into three parts: the selection of parameters for different superpixel algorithms, the selection of different backbones, and the patch-based and superpixel-based comparisons. We introduce them separately in the following subsections.



(a) (b)

Fig. 4. Results of different superpixel segmentation algorithms when the parameters are consistent. (a) The Felz algorithm. (b) The Slic algorithm.

Table 3
Quantitative results of classification and segmentation of Felz algorithm and Slic algorithm with different parameters. In the Felz algorithm, we set the number of segments to 80, 100, 120, 150, 200 respectively, and in the Slic algorithm, we set the number to 100, 200, 250, 300, respectively.

Type	Parameters	Classification			Segmentation		
		Precision	Recall	F_1	DSC	mIoU	HD
Felz	80	0.973	0.951	0.962	0.703	0.380	9.681
	100	0.954	0.935	0.944	0.818	0.530	8.719
	120	0.941	0.938	0.940	0.839	0.579	8.397
	150	**0.998**	0.969	**0.983**	**0.856**	**0.605**	**8.223**
	200	0.966	0.938	0.952	0.812	0.515	8.595
Slic	100	0.911	0.997	0.941	0.670	0.371	9.921
	200	0.933	**1.000**	0.965	0.767	0.467	8.812
	250	0.916	0.990	0.952	0.796	0.492	8.704
	300	0.864	0.997	0.926	0.786	0.482	8.937

4.4.1. Parameter selection of superpixel method

For the selection of superpixel extraction method, we employ the Felz algorithm instead of the more popular Slic algorithm, because it can be in separate under the condition of less area, the border of segmentation more accurate. It can be seen from Fig. 4, that when segmenting the same number of superpixels, Felz segmentation is more refined than Slic. We compare the results of the two algorithms under different parameters, in the Felz algorithm, we set the number of segments to 80, 100, 120, 150, 200, respectively, and in the Slic algorithm, we set the number to 100, 200, 250, 300, respectively. It can be seen in Table 3 and Fig. 5, as the number of superpixels increases, the performance first improves due to the inclusion of more image details, and then decreases after reaching a peak. This is consistent with our intuition, since the assumption that pixels within the same superpixel belong to the same class holds only when the scale of the superpixels is appropriate, and the highest segmentation accuracy is achieved when using Felz to segment the image into 150 blocks.

4.4.2. Effect of different backbones

In common instance-level MIL algorithms, CNNs are typically employed as classifiers. In our method, however, we innovatively utilize ViT as the classifier. Consequently, we conduct experiments targeting various backbones in the MSIL framework. Table 4 shows the effect of the different backbone on segmentation performance. When we use ViT as a classifier, the classification and segmentation results are significantly improved, which indicates that the Transformer's ability to extract features is better than CNNs.

4.4.3. Effect of superpixels

Table 5 shows the results of patch-based and superpixel-based MIL on classification and segmentation. The most important advantage of our methods is its good interpretability and results of segmentation can

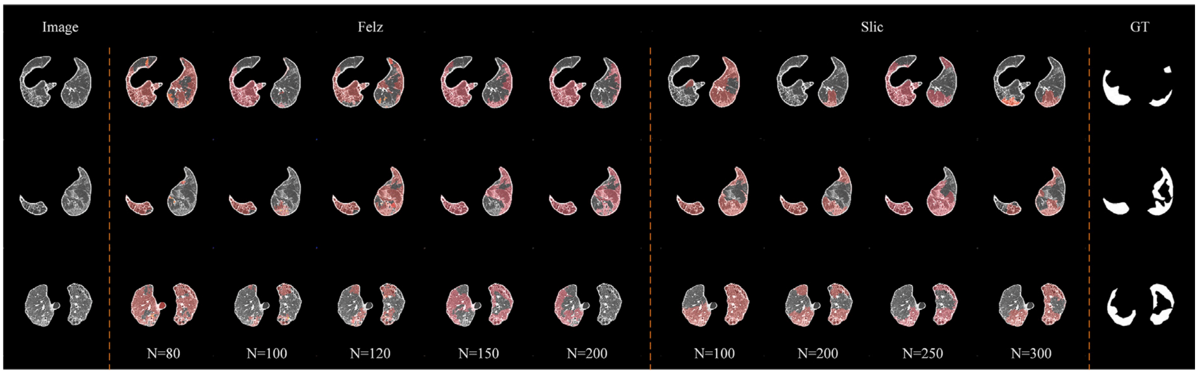


Fig. 5. Visualization of the segmentation results of Felz algorithm and Slic algorithm with different parameters.

Table 4
The results of different backbone in MSIL framework.

Type	Backbone	Classification			Segmentation		
		Precision	Recall	F_1	DSC	mIoU	HD
CNN	ResNet	0.901	**0.974**	0.936	0.664	0.276	9.698
Transformer	ViT	**0.998**	0.969	**0.983**	**0.856**	**0.605**	**8.223**

Table 5
The results of the patch-based and superpixel-based MIL.

Type	Parameters	Classification			Segmentation		
		Precision	Recall	F_1	DSC	mIoU	HD
Patch-based MIL	8 × 8	0.929	0.984	0.955	0.507	0.220	9.698
	16 × 16	0.968	0.994	0.981	0.759	0.449	**7.702**
	32 × 32	0.985	0.997	0.991	0.769	0.441	8.093
	64 × 64	0.991	**1.000**	**0.995**	0.765	0.405	10.219
Superpixel-based MIL	Felz-150	**0.998**	0.969	0.983	**0.856**	**0.605**	8.223

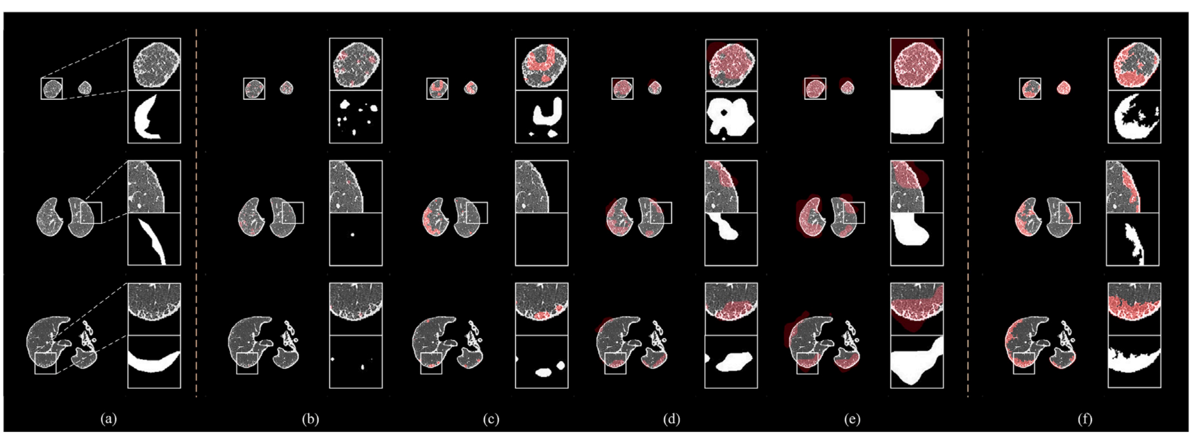


Fig. 6. The results of ablation experiment. (a) The pre-processed image and ground truth; (b) MIL with patch size 8 × 8; (c) MIL with patch size 16 × 16; (d) MIL with patch size 32 × 32; (e) MIL with patch size 64 × 64; (f) MIL with superpixels.

be observed in Fig. 6. It can be seen that although the classification results of patch-based MIL are as good as those superpixel-based, the segmentation effect is unsatisfactory. It can be intuitively seen from Fig. 6 that patches of different sizes have a great impact on the segmentation results, and neither of them can accurately segment the lesion edge. This demonstrates that superpixel can benefit refine boundaries.

4.5. Comparative experiment

In order to demonstrate the efficacy of our method, we conduct a series of comparative experiments. Firstly, we re-implement WSS methods, such as Online Attention Accumulation (OAA) (Jiang, Han, Hou, Cheng, & Wei, 2021) and Group-WSSS (Zhou et al., 2021); in addition, we compare with some current works that apply MIL based on WSS to medical image segmentation, such as DWS-MIL (Jia et al., 2017) and SA-MIL (Li et al., 2023); secondly, recognizing MIL as a classification algorithm (CA), we integrate classical classification algorithms with class activation maps (CAMs) for experimentation, like VGG (Zhou, Khosla, Lapedriza, Oliva, & Torralba, 2016), ResNet (He, Zhang, Ren, & Sun, 2016), Densenet (Huang, Liu, Van Der Maaten, & Weinberger, 2017), these methods can perform WSS based on CAMs to generate a complete mask prediction; finally, we perform experiments on our dataset utilizing fully supervised segmentation (FSS) methods, like Unet (Ronneberger, Fischer, & Brox, 2015), Deeplabv3+ (Chen, Zhu, Papandreou, Schroff, & Adam, 2018) and RDRNet (Lai et al., 2024). For WSS, MIL based WSS and CAMs based CA, we follow the

Table 6
Comparisons with weakly supervision methods, MIL, classification algorithm with CAMs and fully supervision methods. WSS, CA and FSS are denoted to weakly supervised segmentation, classification algorithm and fully supervised segmentation, respectively.

Type	Methods	Classification			Segmentation		
		Precision	Recall	F_1	DSC	mIoU	HD
WSS	OAA (Jiang et al., 2021)	–	–	–	0.372	0.059	22.627
	Group-WSSS (Zhou et al., 2021)	–	–	–	0.554	0.260	13.458
MIL-based WSS	DWS-MIL (Jia et al., 2017)	–	–	–	0.627	0.217	14.330
	SA-MIL (Li et al., 2023)	–	–	–	0.615	0.205	14.650
CAMs-based CA	VGG (Zhou et al., 2016)	0.924	**1.000**	0.961	0.526	0.163	12.955
	ResNet (He et al., 2016)	0.999	0.994	0.991	0.664	0.276	17.390
	Densenet (Huang et al., 2017)	**1.000**	0.997	**0.999**	0.703	0.339	14.704
FSS	Unet (Ronneberger et al., 2015)	–	–	–	0.696	0.350	9.591
	Deeplabv3+ (Chen et al., 2018)	–	–	–	0.724	0.389	9.463
	RDRNet (Lai et al., 2024)	–	–	–	0.811	**0.709**	**3.491**
Ours	ViT-MSIL	0.998	0.969	0.983	**0.856**	0.605	8.223

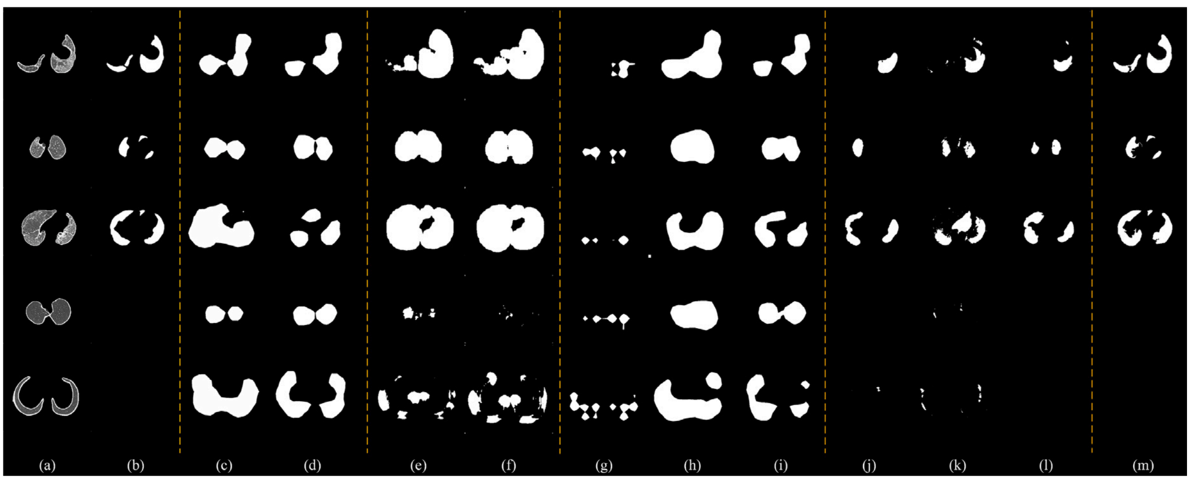


Fig. 7. Visualization results of all methods. (a) Pre-processed image. (b) Ground Truth. (c) OAA. (d) Group-WSSS. (e) DWS-MIL. (f) SA-MIL. (g) VGG-CAMs. (h) Resnet-CAMs. (i) Densenet-CAMs. (j) Deeplabv3+. (k) Unet. (l) RDRNet. (m) Ours. The rows 1-3 are the results on positive images. The row 4-5 are the results on negative images.

standard training and testing settings. For FSS, we use our previous work's dataset with ground truth to train the model.

As shown in Table 6 and Fig. 7, comparing WSS methods relied on image-level annotation (OAA and Group-WSSS), our method significantly outperforms those methods. These methods perform well on natural images because the segmented objects in natural images often have regular shapes and boundaries, but they are not suitable for medical images with blurred lesion edges. Although the MIL method based on WSS (DWS-MIL and SA-MIL) can perform pixel-level segmentation, both of these methods segment cancer areas on pathological images. Due to the obvious difference between pathological images and the CT images we used, these methods are not effective on our dataset. For CAMs based on classification algorithms (VGG, ResNet, Densenet), it can be seen that although these methods perform well in classification evaluation indicators, from the visualization results (columns 7, 8 and 9 of Fig. 7), the ROIs that the method focuses on is not the lesion, but may be the overall lung parenchyma. Due to our pixel-level GT being derived from previous work, only 410 labeled images were available for training the FSS method, which may lead these methods perform poorly. But the result is getting close to our previous work on ILD lesion segmentation (RDRNet).

5. Discussion

With the development of deep learning, researchers have developed a variety of deep learning-based algorithms to help the classification of ILD and the segmentation of lesions. But these segmentation methods are fully supervised learning, however, obtaining these fine-grained annotations is usually time-consuming and labor-intensive for doctors. The WSL could solves the above difficulties.

In this work, we propose a novel WSS framework, proposing to combine superpixels, Transformers, and MIL to accurately segment the lesions of ILD with only image-level labels. We demonstrate the effectiveness of our proposed method through extensive experiments. The results of comparative experiments show that our method outperforms the current WSS methods and FSS methods, and is getting close to our previous work on ILD lesion segmentation (RDRNet).

Effect of Transformer: We compare the performance of CNNs module and Transformer module as backbone for MSIL. Specifically, in this study, we select ResNet as CNN architecture and ViT as the Transformer architecture. From the results, we can see that using ViT as the classifier improves the classification effect by 0.097 and the segmentation effect by 0.192 compared with ResNet. This shows that the self-attention mechanism in Transformer can better comprehend global contextual information, which helps enhance the overall semantic understanding of the image, thus segmentation more accurately.

Effect of superpixel: Most MIL algorithms are based on patches or pixels, and we have also conducted experiments on this. We conduct experiments based on patch sizes of 8 × 8, 16 × 16, 32 × 32 and 64 × 64, respectively. It can be seen from the results that since the size of the patch is a fixed rectangle, the edge identification of the lesion is not accurate. Although from the quantitative results, the accuracy increases with the increase in size, from the visual results, the segmentation edges are not accurate. This is because the larger the size, the more lesions are included, so the quantitative results are better. Since superpixel is a small region composed of a series of adjacent pixels with similar features such as color, brightness and texture, it not only retains the effective information of image segmentation, but also does not destroy the boundary information of ROIs in the image. Therefore, taking superpixel as an instance, it can segment the lesion boundary better than patch-level, and reduce the complexity of image processing than pixel-level.

6. Conclusion

In this work, we propose a novel WSS model for ILD lesion segmentation, and we explore the capability of MIL for medical image segmentation based on the superpixel instances. In order to solve the problem of blurred boundary of the lesion and the independency of instances, we innovatively propose the ViT-MSIL framework. Specifically, we introduce superpixels into MIL and use superpixels instead of patches as instance. From the results, the propose MSIL can segment the lesion edge more accurately. In addition, we adapt the ViT into MSIL pipeline, which can extract long-range dependencies of all the embedded entities from a more global perspective. Extensive experiments are conducted on our dataset for ILD lesion segmentation, and demonstrate that our proposed ViT-MSIL outperform both WSS methods, CAMs based methods and FSS methods. In future work, we hope to refine the types of lesions and realize the segmentation of multiple lesions, and the continuity of HRCT data is used to realize the 3D segmentation of lesions.

CRediT authorship contribution statement

Yexin Lai: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Formal analysis, Writing – original draft, Data curation, Visualization. **Xueyu Liu:** Methodology, Software. **Linning E.:** Supervision, Data curation. **Yujing Cheng:** Resources, Data curation. **Shuyan Liu:** Resources, Data curation. **Yongfei Wu:** Supervision, Writing – review & editing, Funding acquisition. **Wen Zheng:** Supervision, Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declared that they have no conflicts of interest to this work. We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

Data availability

The data that has been used is confidential.

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